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## Chapter 9

### Distribution Requirements Planning Overview



# Introduction

Distribution Requirements Planning provides the necessary data for matching customer demand with the supply of products at various stages in the physical distribution system.

DRP also helps management to anticipate future requirements in the field, more effectively deploy inventory to meet customer-service requirements, and more rapidly adjust to the vagaries of the market-place.

DRP evaluates the demand at the Distribution Centres (DCs) and generates supply plans for centres at a product level. These time-phased plans, or schedules, provide the information planners need to match supply and demand.

The planner can then confirm suggested supply orders, check the feasibility of suggested changes, monitor data integrity and assess the impact of new situations.

## Relationship To Other Modules

DRP is one of the Advanced Application Modules which extend the functionality of Base Application Modules.

The applications which are pre-requisite for the operation of DRP are:

- Inventory Management
- Sales Order Processing
- Purchase Management
- Requisitioning
- General Ledger
- Accounts Receivable
- Accounts Payable

DRP also interfaces with, and enhances, Warehousing, Production, World Trade, and Forecasting.

At DCs, when you confirm shipment receipt, you can enter warehousing details to include the items in the warehousing put-away processing.

At the Supply Centres, you can pick items by Distribution Order, within Warehousing. A Distribution Order is an internal order for transferring goods from Centre to DC, DC to DC or DC to Centre. It is generated from a DRP suggestion or entered manually, and is handled as an internal order within Sales Order Processing.

If the Supply Centre is a Production site, the Distribution Orders are considered as demand within the Production systems MPS/MRP processing runs. (To achieve this define items at the Centre to Production).

If World Trade is attached, you can add or amend Distribution Orders in both Distribution Requirements Planning and World Trade (only DRP files are updated). When you transfer of goods, which cross an EC Border (via order processing/warehousing confirmation of despatch), movement records are written for processing in World Trade and are used to produce Intrastat and ESL reports, and to post the acquisition VAT due on receipt of the goods.

You can transfer sales forecasts from Forecasting into any DRP Centre or DC, at item or minor item group level.

## DRP Configuration

You can operate Distribution Requirements Planning for a number of companies. You maintain the characteristics of each company on control files.

DRP uses the control data stored for Inventory Management. However, you need to enter further information at company level specifically for DRP.

You can control the following DRP policies at each company:

- Method of forecast consumption.
- Maximum number of lines per distribution order - that is a supply order from Centre to DC. You can change this when you create orders from DRP suggestions.
- Default calendar parameters.
- Percentages of quotations to include DRP runs (only if Advanced Order Entry is installed).
- Default absolute forecast error tolerance percentage.
- Calculation of safety stocks.
- Period forecasting.

## Consumption of Forecast

You can assess demand by comparing the sales forecasts with actual sales orders by period and to plan for the greatest.

Also you can assess demand using the Consumption of Forecast. This compares cumulative forecasts with cumulative demand. This is because if a period demand exceeds the period forecast it may be because demand forecast for a following or prior period had either arrived early or late. That is the quantity forecast is accurate but the period due is not. Consider the following demand:

| <b><u>Period Start</u></b> | <b><u>1 Mar</u></b> | <b><u>8 Mar</u></b> | <b><u>15 Mar</u></b> | <b><u>22 Mar</u></b> | <b><u>Total</u></b> |
|----------------------------|---------------------|---------------------|----------------------|----------------------|---------------------|
| Forecast                   | 100                 | 100                 | 100                  | 100                  | 400                 |
| Actual Sales               | 150                 | 90                  | 30                   | 110                  | 380                 |
| Demand A                   | 150                 | 90                  | 60                   | 100                  | 400                 |
| Demand B                   | 150                 | 100                 | 100                  | 110                  | 460                 |

Where Demand A uses forecast consumption, and B does not.

There are three modes of Forecast Consumption:

- Consumption throughout a DRP run without reference to period ends.
- Consumption of forecast over individual periods.
- Forecast consumption over individual weeks.

These generate different demands for re-supply which mean different opening and closing stock positions, affecting stock holding costs and customer service.

The mechanics of consumption of forecast are:

1. Sum the forecast to date.
2. Sum the actual sales to date.
3. Set the demand as the greater of forecast and actual sales.
4. Sum the consumed demand to date, excluding the current period.
5. Calculate the consumed demand for the current period as Demand to date (from 3) - Consumed Demand to date (from 4).

Reconsidering the previous example, then:

| <b><u>Period Start</u></b> | <b><u>1 Mar</u></b> | <b><u>8 Mar</u></b> | <b><u>15 Mar</u></b> | <b><u>22 Mar</u></b> | <b><u>Total</u></b> |
|----------------------------|---------------------|---------------------|----------------------|----------------------|---------------------|
| Forecast                   | 100                 | 100                 | 100                  | 100                  | 400                 |
| Actual Sales               | 150                 | 90                  | 30                   | 110                  | 380                 |
| Forecast to date           | 100                 | 200                 | 300                  | 400                  |                     |
| Actual Sales to date       | 150                 | 240                 | 270                  | 380                  |                     |
| Demand to date             | 150                 | 240                 | 300                  | 400                  |                     |
| Consumed Demand            | 150                 | 90                  | 60                   | 100                  | 400                 |

## Calendar Parameters

You can enter a default calendar code for the company. DRP planning runs use this for any Centre or DC which does not have a calendar assigned to it.

You can enter the following period details:

- Number of periods per year.
- Period Names.
- Number of days per period.

The DRP Calendar Maintenance function uses these when generating a calendar.

You can also maintain a standard week template. This specifies the normal working week. DRP Calendar Maintenance displays and uses this when generating calendars.



*Note: The DRP planning runs suggest actions only on working days.*

## Maintenance of Reference Data

Before you use Distribution Requirements Planning you must plan and prepare carefully.

Consider the supply chain network and flow of products between them, particularly the relationship between distribution centres and stockrooms. For planning purposes, you can associate multiple stockrooms with a distribution centre yet for any item only define one stockroom as the primary receiving and issuing store.

As this is an advanced application module, you set up much of the primary data as part of the base modules. However, you must enter additional information to get the maximum benefit from DRP.

## Reference Information Required

You need to maintain Information for the following:

### **Company**

This contains control and parameter information. You must set up the company in Inventory Management. It defines the default calendar, maximum number of lines on the Distribution Order, whether you require consumption of forecast, quotation percentage (if Advanced Order Entry is installed) and forecast error tolerance.

### **Calendars**

This defines, by year, alternative calendars used for planning. Different Centres or DCs and seasonal profiles can use different calendars. You can enter a number of calendars per company. For each calendar, you can maintain up to five week templates specifying the working days in a normal week, and the dates to which it applies.

### **Periods**

This defines for a Centre, or DC, the forecast periods used for planning. You maintain the period intervals for the planning runs per Centre or DC. You can specify the intervals by calendar days. You can specify up to 46 periods. The DRP planning runs assess the supply and demand in each period, beginning at the start date for the run.

### **Stockrooms**

This defines the stockrooms associated with a site and controls which stock balances, demands, and supply orders DRP considers during a planning run. You can define any number of Inventory Stockrooms to a single Centre or DC. You must have defined the stockroom to Inventory Management and specified periods for the Centre / DC. You must assign each stockroom an SOP dummy internal customer number, to reflect distribution orders raised for that stockroom.

If there is only one stockroom per site, you can also assign the stockroom code to the Centre or DC. You can only assign a stockroom to one Centre or DC.

For multiple stockrooms, the DRP planning runs extract stock, supply and demand information for all stockrooms at the selected DC or Centre.

### **Customers**

This defines internal customer codes associated with stockrooms for distribution centres to generate order details and despatch documentation that support stock transfers between Centres / DCs. You must enter customer numbers for each DRP defined stockroom. These are valid customers from within Sales Order Processing with a delivery sequence code of 001. These customers hold the internal distribution orders raised for supply to that particular stockroom.

### **Supply Networks**

This defines the supply chain planning relationship. You must specify the relationship of Centres or DCs within a network. You must have defined periods and stockrooms for each Centre / DC.



You can define multiple networks for a company and include sites in more than one network, normally as a Centre in one level and a DC in another. You must set up a supply network for each level of supply, normally, the Centre of one network is a DC, supplied by an alternative Centre, within the other network. You can define a site to any number of networks, using the same DC or Centre.

Simply give a network a code, description, Centre, and all DCs supplied by this centre.

### **Planner Codes**

This defines the person responsible for planning an item or range of items. You must maintain at each centre and DC each planner who will use DRP. You enter personal details and a cross reference to the purchase officer number in Purchase Management.

You assign this planner code to items, for which the planner is responsible, in the item master details.

### **Item Group Profiles and Forecasts**

For each item group for any centre or DC you can define the percentage sales of each item and maintain sales and stock forecasts by value or quantity. You enter an annual forecast spread over the weeks in the specified year.

### **Item Sales Forecast and Stock Levels**

You can generate Item Sales Forecasts from the item group forecasts and profile. You can maintain the forecast manually.

You can generate Item Stock Levels from the item group profile and target stock quantity and range of dates. You can maintain these manually.

### **Net Change Reason Codes**

This controls which operational transactions set net change processing flags and affect the efficiency and run times of DRP processing in net change mode.

### **Supplier Schedule Profile**

This controls the format and scope of vendor supply schedule reports.

### **Seasonal Profiles**

you can define period, weekly and daily indices which you can use to spread sales forecasts and stock build requirements interactively and automatically.

### **Processing Profiles**

To update inventory when you despatch and receive distribution orders, you must set up the DRP Processing Profile record in Inventory Management. Refer to the Utilities section of the Inventory Management Product Information for further details.

## Item Master Details

This defines by item and Centre / DC the re-supply rules used for planning. You must define items to the local stockroom within Inventory Management. You must define the Centre or DC to at least one network.

You do not need to define an order policy at the centre for a centrally supplied item unless the item is included in the DRP run for the Centre.

For DRP to control how its supply is administered you must enter the following:

- How much is ordered when a net requirement has been calculated?
- Where is the main source of supply?

To satisfy the first requirement specify an order policy and the appropriate quantities. The choice is as follows:

| Order Policy | Description                                                                                                                  | Mandatory Quantities                      |
|--------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| A            | Discrete. Order the quantity required.                                                                                       | None.                                     |
| B            | Discrete above minimum. Order quantity required if greater than minimum. Otherwise use minimum quantity.                     | Minimum order quantity.                   |
| D            | Fixed quantity. Generate multiple orders of fixed quantity to satisfy the required quantity.                                 | Fixed order quantity.                     |
| G            | Number of days supply. Accumulate the requirements over specified number of days and suggest an order to satisfy the demand. | Number of days supply.                    |
| H            | Fixed quantity above minimum. Order quantity must be greater than minimum and divisible by the multiple quantity.            | Minimum order qty.<br>Multiple order qty. |

Satisfy the second requirement by specifying whether you purchase the item locally or centrally or manufacture it centrally. If you supply the item centrally, you must enter a centre and stockroom defined to the Centre. DRP then creates a distribution order line for this central stockroom.

You must enter item master details for the item at each DC it is sold from. If you centrally supply the item, you must also enter details at the supply centre.



*Note: If the supply centre is a production site, define these items to the Production System and plan them via MPS (after evaluating the DC demand and passing demand to the centre as distribution orders).*

Otherwise, define the items at the centre to DRP and the central supply plan created within it.

## Starting to use DRP

When you have set up the basic data you can start using DRP.

The system can access data maintained through other modules; for example, from Inventory Management - stock; from Sales Order Processing - outstanding sales orders quotations (if AOE installed); from Purchase Management - supply from external sources; from Forecasting - Item and Item Group sales forecasts.

If there are outstanding transfer orders between distribution centres, you must enter the details manually using the ad hoc Distribution Order Maintenance functions. If the number of orders are significant and on an existing computer system, it may be more cost effective to transfer data using a conversion routine commissioned for the purpose.

## Implementation Procedure

After installation, the recommended implementation steps for DRP are:

1. Within Sales Order Processing, create customer records for each stockroom, with a delivery address number of 001.
2. Make sure the necessary processing profile records are in Inventory Management. Refer to Inventory Management Product Information (Utilities section) for further details.
3. Create the DRP company profile record, using the same company code as the one defined to Inventory Management.
4. Create the DRP calendars for this year, and any subsequent years.
5. Design the DRP networks, and how / when they are processed.
6. Create the DRP period records for each centre and DC.
7. Create the DRP stockroom records for each centre and DC.
8. Assign the internal customer number to the stockroom within DRP Customer Maintenance.
9. Create the DRP network and planner records.
10. Define the Item to all stockrooms at centres and DCs via Inventory Management.
11. Create the distribution item master records for every centre or DC which stocks or sells the item.
12. Create item group profiles and forecasts.
13. Create seasonal profiles.
14. Generate item sales forecasts.
15. Maintain the generated sales forecasts, where appropriate.
16. If you require stock building, enter Item group stock level parameters and create item stock levels.
17. Maintain the generated item stock level forecast, where appropriate.
18. If you are using net change DRP, set up net change reason codes.
19. If you require the supplier schedule report, define supplier schedule profiles.



*Note: If you only create sales forecasts in Forecasting, you do not need to do steps 12 to 14. However, if you maintain seasonal profiles then Forecasting applies them when you transfer into DRP forecasts.*



*Note: You can run DRP based on actual demand only, that is excluding forecasts*

Only steps 1 to 11 are mandatory.

# Forecast Manipulation

You can enter Sales Forecasts at item group (minor) level and then spread, on a time phased basis, over defined time bands down to period, weekly and daily levels in stages either by:

- Interactively selecting definable seasonal profiles.
- Invoking automatic spreading.
- Direct entry and manipulation of forecasts and spread indices.

You can maintain the percentage spread of sales amongst items within an item group. This is known as the item group profile.

From the item group profile and item group sales forecasts, you can automatically generate and then maintain time phased item level sales forecasts.

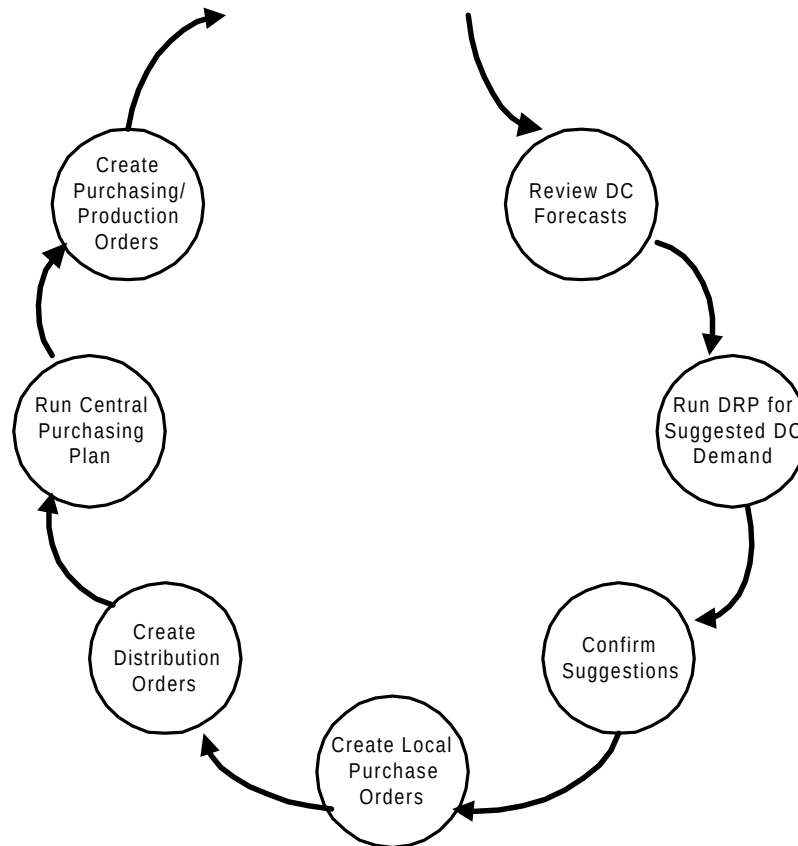
You can also build item level forecasts directly, even if the items do not belong to a defined item group or are not included in the group level forecast.

You can transfer sales forecasts into DRP from Forecasting. This avoids entering manual forecasts and takes into account the last three years' sales demand.

You can also enter stock forecasts to specify stock building requirements. Creating demand, in addition to sales, to build stocks covering future periods when supply would be constrained and insufficient. Typically, this is where there is a highly seasonal demand and a belief that suppliers will not have the capacity to meet the requirements unless they are smoothed and pulled forward.

# Processing

## The DRP Cycle



- Review DC Forecasts - Use DRP Forecast Maintenance options to review and update the stock and sales forecasts for relevant items.
- Evaluate DC Demand - DRP, via DRP processing run, assesses the supply and demand figures for all DCs in a specified network, suggests new supply orders and amendments to existing supply orders, and deletes all previous supply suggestions.
- Confirm Suggestions - Review and accept the suggestions, then use Requisitioning to convert suggestions to requisitions. You can amend these within Requisitioning before creating purchase or distribution orders.



*Note: All outstanding requisitions sourced from DRP for the DC stockrooms are discarded and new ones added from the latest DRP suggested orders for the DC.*

- Create Local Purchase Orders - Within Requisitioning, convert the purchase requisitions created from DRP into purchase orders on the relevant external suppliers. Transfer the purchase into Purchase Management.
- Create Distribution Orders - Within Requisitioning, convert the distribution requisitions, for supply from a centre, into distribution orders. Transfer the

distribution orders into Sales Order Processing. The order lines are placed against the centre stockroom defined to the item in DRP. You can also enter distribution orders directly into Sales Order Processing via distribution order entry or amend. Specifying the from and to stockrooms, to automatically create the link between the two stockrooms. This creates an internal sales order for the customer defined for the stockroom which receives the supply.



*Note: The distribution order is an internal sales order, so no invoice details are raised. The customer is the one defined to the stockroom which will receive the supply.*

*When you confirm despatch it adds the distribution order lines to the DRP shipments in-transit file. If World Trade is attached and the transfer is across an EC border, it creates a movement record for World Trade to use to produce Intrastat and ESL reports, and to post the acquisition VAT due on receipt of the goods.*

*When you receive distribution order lines at the stockroom, confirm the goods as received via DRP processing.*

- Run Central Supply Plan - Assesses the supply and demand figures for the selected supply centre, suggests new purchase orders from external suppliers and recommends amendments to purchase orders.



*Note: If the centre is a production site, you must define the items to production not DRP, and the centre run should be an MPS run not a DRP one. Otherwise, this centre run is self-contained within DRP.*

- Create Purchasing/Production Orders - If the centre run was within DRP, create the requisitions from DRP suggestions and subsequently make into purchase orders using Requisitioning. If the centre run was within Production, create the purchase requisitions and then convert into purchase orders, using Requisitioning. Also, you can create production orders from MPS suggestions within Production Master Production Scheduling.

## Run Processing Options

In Distribution Requirements Planning you can select the items for re-planning in the DRP run processing options (DC, Centre and Network planning runs) in a number of ways.

There are three distinct run types as follows:

### **1 - Full regeneration**

Re-plans all DRP items for the selected network or centre.

### **2 - Net Change**

Selects a subset of items for re-planning on the basis of changes in the supply, demand, stock and/or replenishment parameters since they were last planned. This is achieved using net change triggers.

The Distribution Management applications set triggers using a set of defined reason codes.

You can define a net change reason as a type of business transaction which potentially affects the supply, demand, stock and/or replenishment parameters of a particular item.

The reason codes currently set up are:

| Code | Description                     | Applications Where Triggers Set         |
|------|---------------------------------|-----------------------------------------|
| 01   | Forecast vs. Actual Comparison  | DRP                                     |
| 02   | Forecast Generation             | Forecasting                             |
| 03   | Forecast Maintenance            | DRP                                     |
| 04   | Distribution item Master        | DRP                                     |
| 05   | Unplanned Inventory Transaction | Inventory Management                    |
| 06   | Requisition Maintenance         | Requisitioning                          |
| 07   | Unplanned Issue or Receipt      | Inventory Management/Warehousing        |
| 08   | Purchase Order Maintenance      | Purchase Order Management               |
| 09   | Distribution Order Maintenance  | Sales Order Processing / Requisitioning |

### **Example**

You raise a purchase order in Purchase Management for replenishment of item I1 at stockroom S1.



If stockroom S1 is attached to a centre or DC in DRP, and item I1 is defined as a DRP item at that centre or DC, then a net change trigger is written for that particular location / item / reason code combination (provided that trigger has not already been set).

When you review net change triggers for a centre or DC before submitting a net change planning run, the number of triggers set for each reason code is displayed. Use this information to select whether you process some or all of the items which have set triggers under any of the defined reason codes.



*Note: When a particular item is re-planned at a particular DRP location (centre or DC), all net change triggers are deleted for that location / item combination. This is irrespective of the DRP run type chosen for processing.*

### **3 - Selective Processing**

Selects a subset of items for re-planning on the basis of the selection of ranges of DRP items, DRP Planners, Inventory minor item groups, Inventory cost / usage elements and / or preferred suppliers.

You can make these selections before submitting any of the DRP Planning Runs (DC, Centre or Network runs).

An item will only be re-planned at a particular centre or DC under the following conditions:

- It is within the range of items selected, *and*
- The DRP planner assigned to the item is within the range of DRP planners selected, *and*
- The Inventory minor item group to which the item belongs is within the range of Inventory minor item groups selected, *and*
- The Inventory cost / usage element calculated for the item is within the range of cost / usage elements selected, *and*
- The preferred supplier defined on the Inventory item / stockroom details is within the range of preferred suppliers selected.

## DRP Cycle Summary

This represents a one-level DRP cycle run. To accommodate multi-level, you can repeat or partially repeat this cycle, for each required network. You can define a supply centre in one network as a DC in another network.

## Confirm Shipment Receipt

This is the day-to-day functional part of DRP, as distinct from the planning runs.

When you confirm distribution order lines as despatched from supply stockrooms, they are added to a shipments in-transit file. At this point they are considered to be in-transit between the two stockrooms.

On receiving the shipment, you confirm receipt via the relevant DRP processing option. This allows you to receive all order lines or selectively receive lines, with or without amendments.

You must confirm the quantity of items recorded as despatched from the From stockroom as received. You can do this across any number of shipment receipts until you confirm the full quantity. If there is a discrepancy, you must receive the full despatched quantity and do an Inventory stock adjustment, with an appropriate reason for the difference.

When you confirm receipt, the details of each shipment received is output to a report.

You can enter warehousing details if the receiving stockroom is defined as a warehouse to Warehousing.

The Inventory stock figures are updated, and movement records created. This transaction is regarded as a transfer in to the receiving stockroom. Similarly, distribution order lines confirmed as despatched at the supply stockroom are treated as transfers out (within history and movement records).

On receiving an item, the costs are updated, based on the cost held on the in-transit record in the Inventory Management application.

# Enquiries

You can group enquiries into one of three categories:

- Planning - What is to be done?
- Static Information - What are the parameters?
- Status - Where are things?

## **Planning**

You can view all supply and demand information generated by any of the DRP planning runs via the schedule enquiries.

There are separate, almost identical, DC and Centre schedule enquiries. This allows for access restriction, if required.

## **Static Information**

You can enquire on the ordering policies and parameters for each item, at any selected Centre or DC.

## **Status**

You can view the current stock situation for any item / stockroom combination, via the Inventory Management Enquiries.

Use the Shipments in-transit enquiry to view the items in transit between stockrooms. This gives all distribution order lines despatched by the supplying stockroom but not yet confirmed as received at the receiving stockroom.

# Reports

You can enter selection and sequencing parameters before running a report.

Standard reports print on standard listing paper with 132 print positions and 6 lines to the inch. The reports print the procedure or program number on the top left and the page number and date of run on the top right of the report.



*Note: The schedule and valuation reports both have 198 print positions.*

As with the enquiries, you can group the reports into Planning, Static Information and Status.